and generalized treatment are now optimized through data-driven, targeted interventions enabled by UAVs. This shift reduces labor requirements, minimizes fertilizer and pesticide usage through precise application, and improves irrigation efficiency by identifying moisture stress at specific field locations. As a result, farmers achieve higher crop yields, reduced input costs, and improved environmental stewardship, demonstrating that UAV integration supports both *economic viability* and *sustainable food production*, particularly in large-scale and resource-sensitive agricultural systems.

Table 8.5 Economic and sustainability gains enabled by UAV agriculture.↵

Impact area	Traditional farming	UAV-assisted precision farming	Resulting benefit
Monitoring labor	Manual scouting required	Automated aerial scanning	Lower labor demand and faster coverage
Fertilizer and pesticide use	Uniform application	Targeted “apply where needed” precision spraying	Reduced chemical costs and soil preservation
Irrigation efficiency	Broad-area irrigation	Real-time moisture mapping and selective irrigation	Water savings and sustainability
Crop yield	Reactive crop management	Proactive and predictive crop health response	Higher yield and stable production output

8.4.3 Economic efficiency and sustainable food production

UAV-based agriculture delivers substantial economic benefits by reducing input costs, minimizing labor-dependent operations, and improving crop quality and yield. Farmers save resources by applying fertilizer, pesticides, and water only where necessary, while UAV-enabled monitoring reduces